WOOD: InstructPix2Pix White-box Geometry Results

Combined differentiable perturbation results

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WOOD optimizes Z with loss = -Z. Model weights are frozen; only differentiable perturbation parameters are optimized.

Run matrix

model	code objective	objective	cases	iterations	status
InstructPix2Pix	vae_conditioning	VAE conditioning latent	4.0000	150.00	done
InstructPix2Pix	unet_prediction	UNet denoising prediction	4.0000	150.00	done

Aggregate summary

model	objective	runs	mean final Z	mean SSIM	output SSIM	output L2
InstructPix2Pix	VAE conditioning latent	4.0000	150.47	0.5675	0.4973	0.2854
InstructPix2Pix	UNet denoising prediction	4.0000	0.0030	0.7988	0.6209	0.1778

Per-run final values

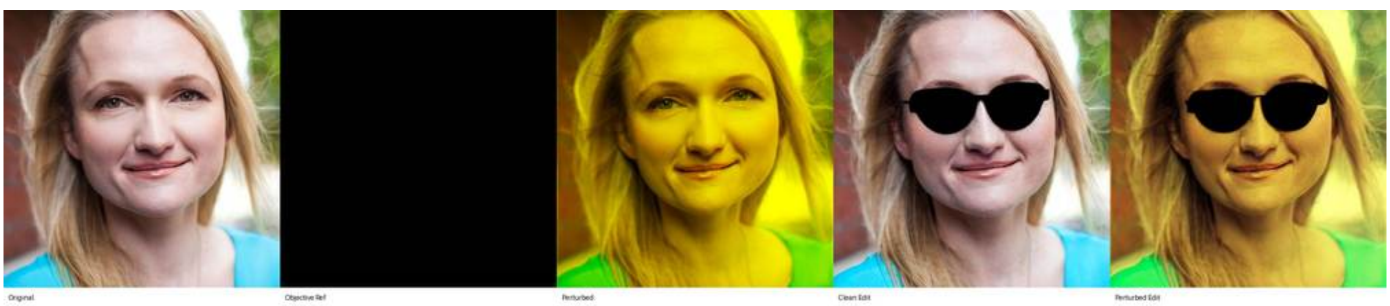
objective	face	prompt	Z	loss	SSIM	out SSIM	out L2	max disp
VAE conditioning latent	face_002	add black sunglasses	148.57	-148.57	0.6374	0.4627	0.2880	14.236
VAE conditioning latent	face_002	add headphones	149.34	-149.34	0.6332	0.4175	0.2996	17.127
VAE conditioning latent	face_005	add black sunglasses	151.99	-151.99	0.5001	0.6479	0.2107	26.987
VAE conditioning latent	face_005	add headphones	151.98	-151.98	0.4992	0.4611	0.3433	24.529
UNet denoising prediction	face_002	add black sunglasses	0.0028	-0.0028	0.6210	0.4207	0.2788	5.6732
UNet denoising prediction	face_002	add headphones	0.0028	-0.0028	0.6304	0.3993	0.3013	6.5670
UNet denoising prediction	face_005	add black sunglasses	0.0031	-0.0031	0.9707	0.8473	0.0603	5.7349
UNet denoising prediction	face_005	add headphones	0.0032	-0.0032	0.9732	0.8163	0.0708	5.6206

Image strips

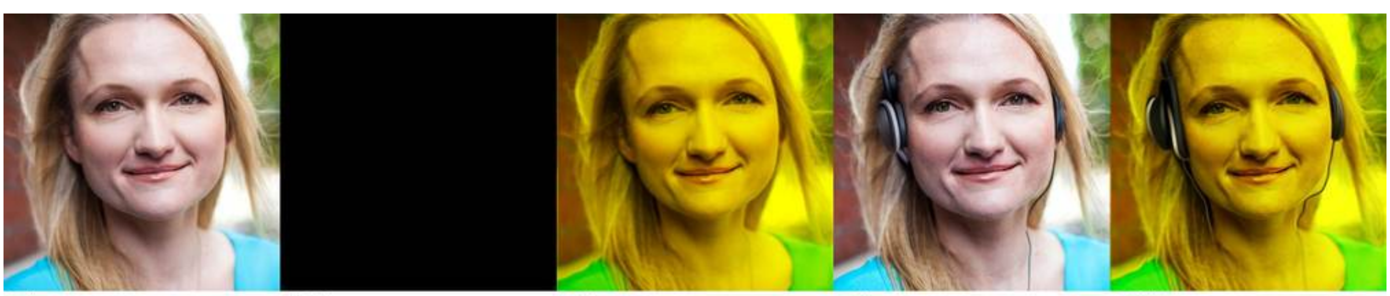
VAE conditioning latent / face_002 / add black sunglasses

VAE conditioning latent / face_002 / add headphones

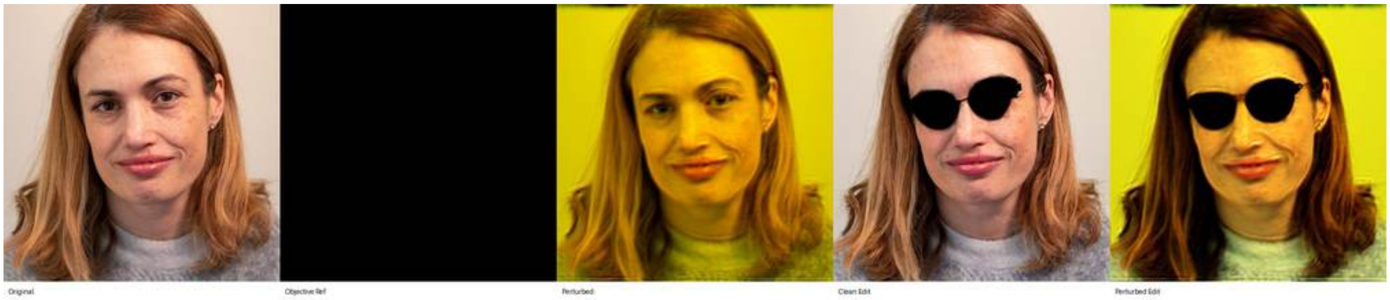
VAE conditioning latent / face_005 / add black sunglasses



VAE conditioning latent / face_005 / add headphones

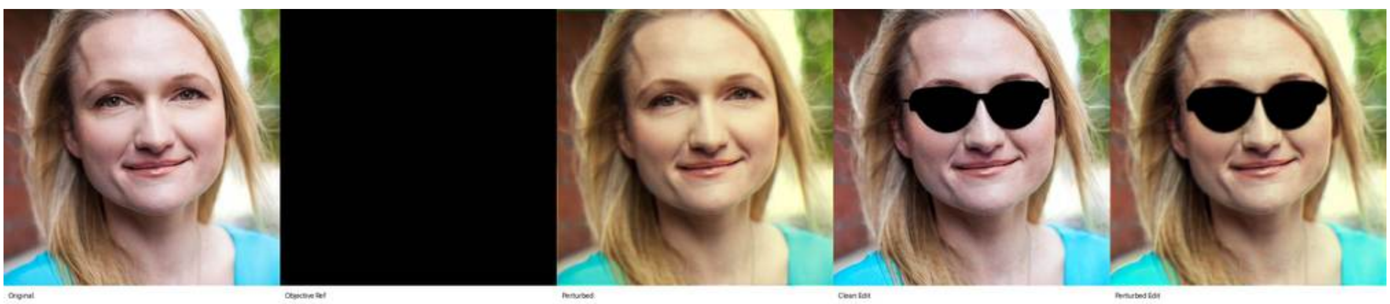


UNet denoising prediction / face_002 / add black sunglasses

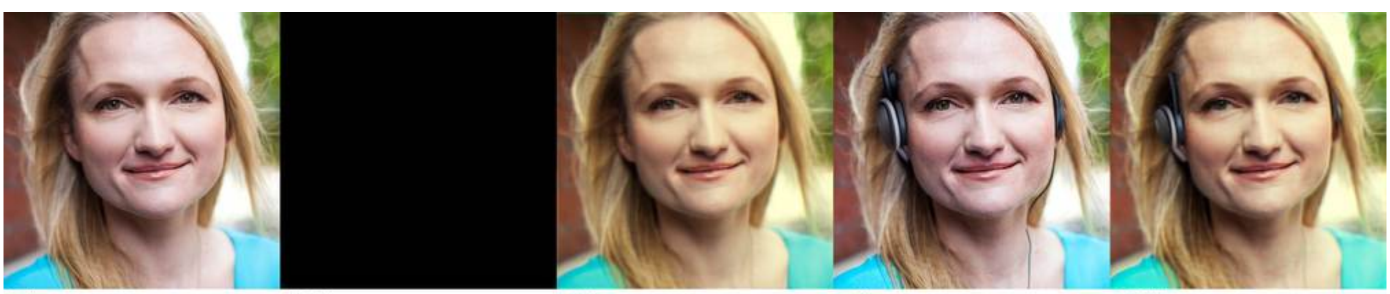


UNet denoising prediction / face_002 / add headphones

UNet denoising prediction / face_005 / add black sunglasses



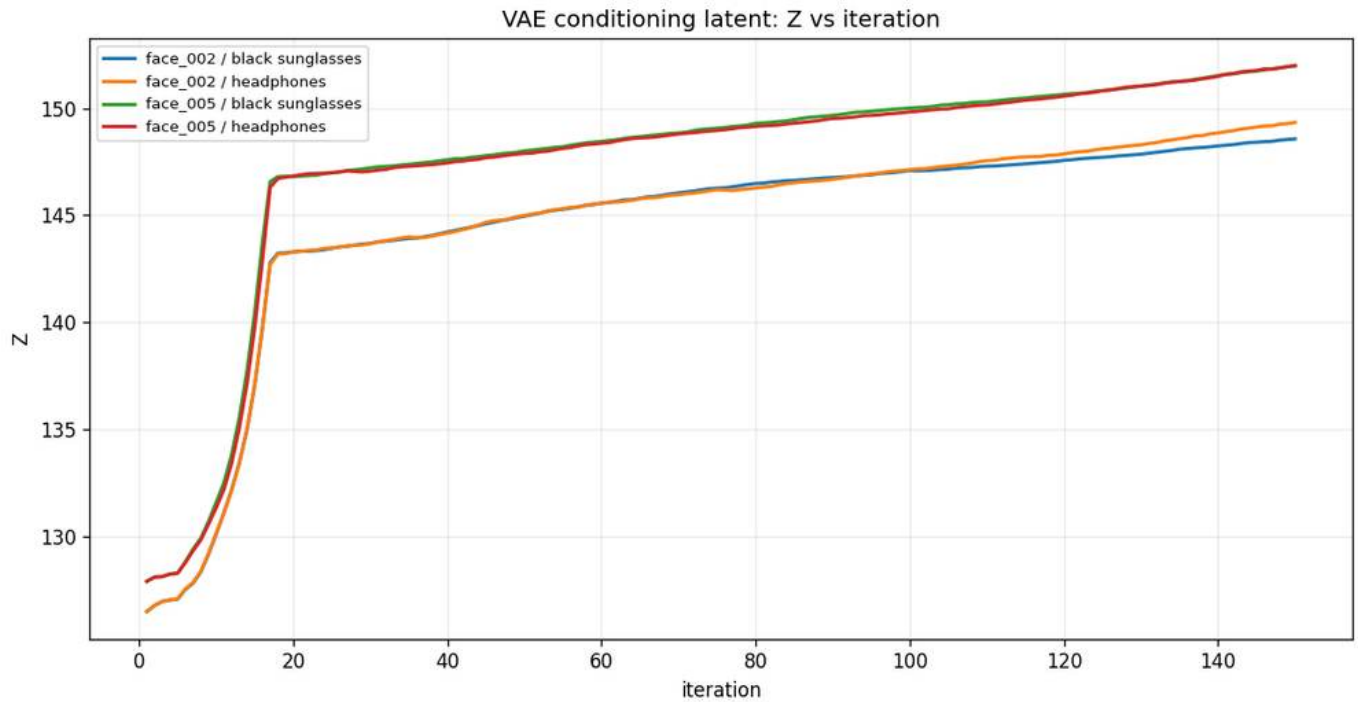
UNet denoising prediction / face_005 / add headphones



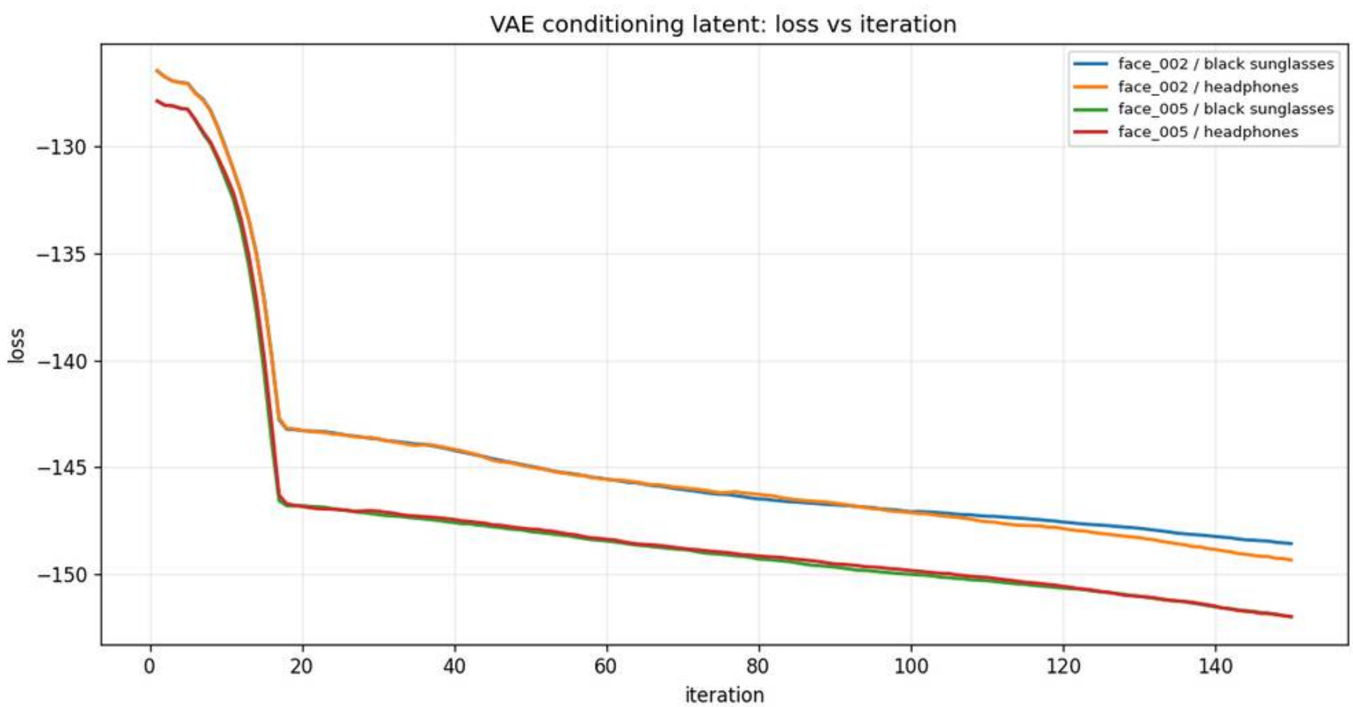
Graphs

VAE conditioning latent

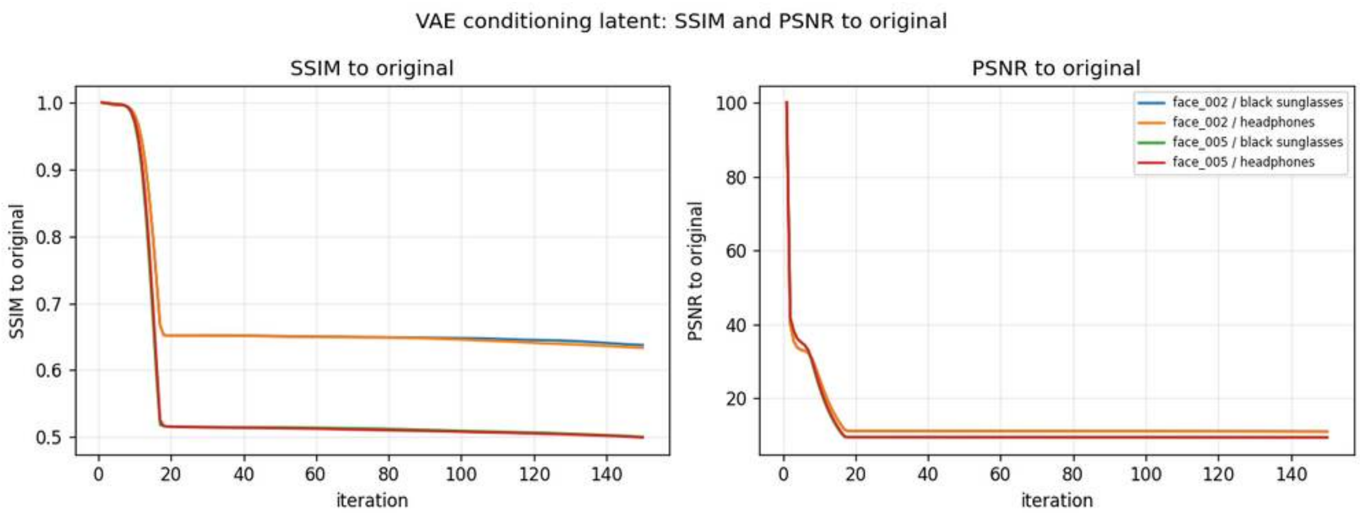
VAE conditioning latent: Z vs iteration



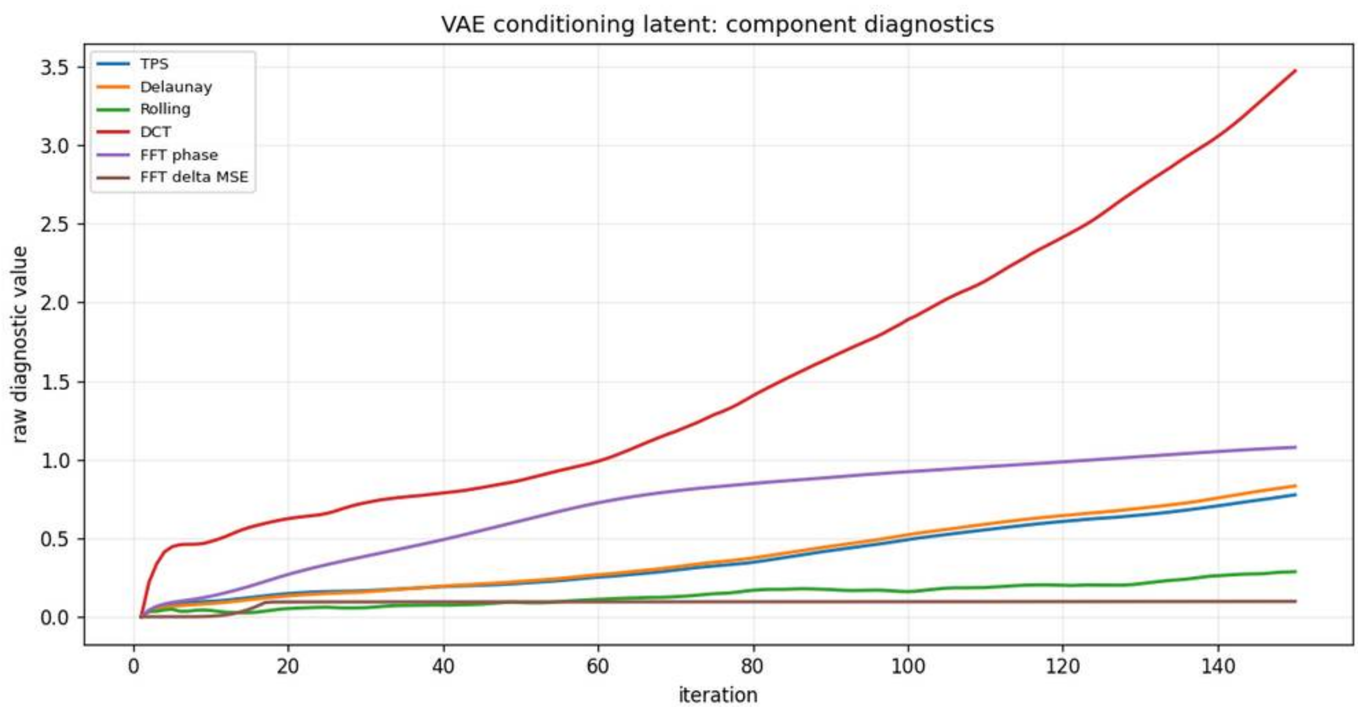
VAE conditioning latent: loss vs iteration



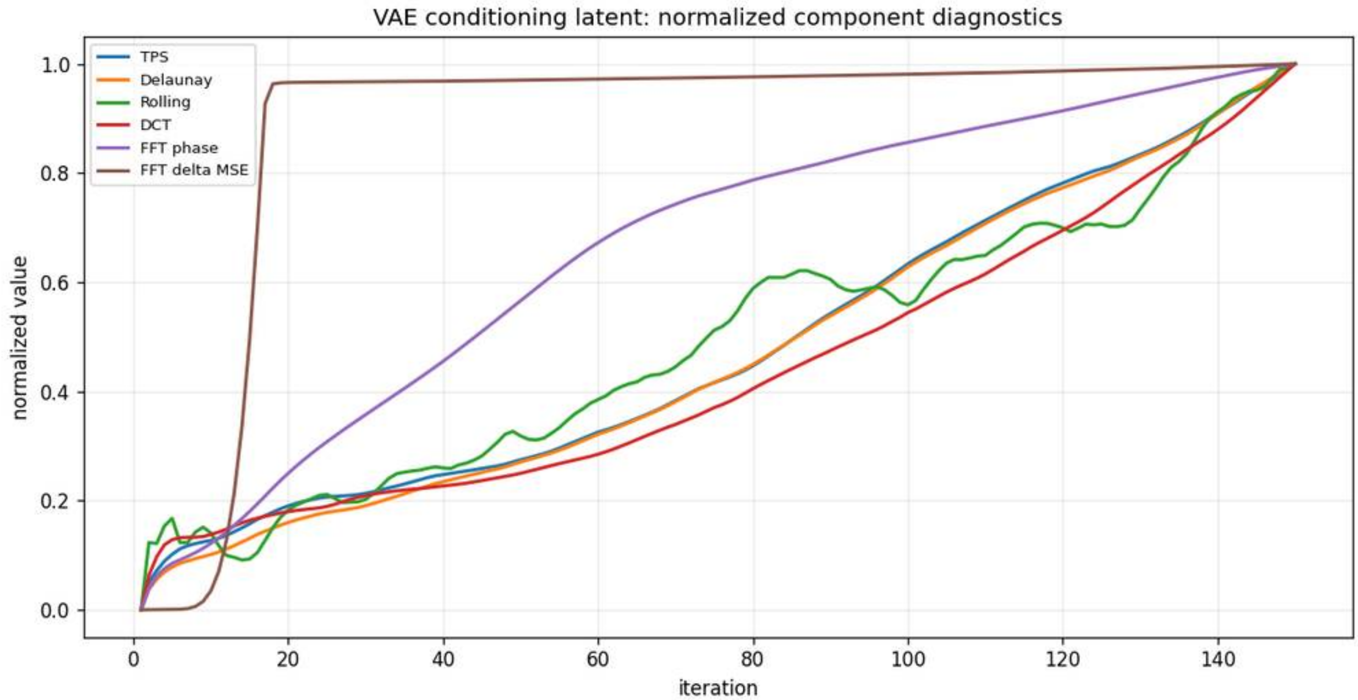
SSIM and PSNR to original



Geometry component contribution

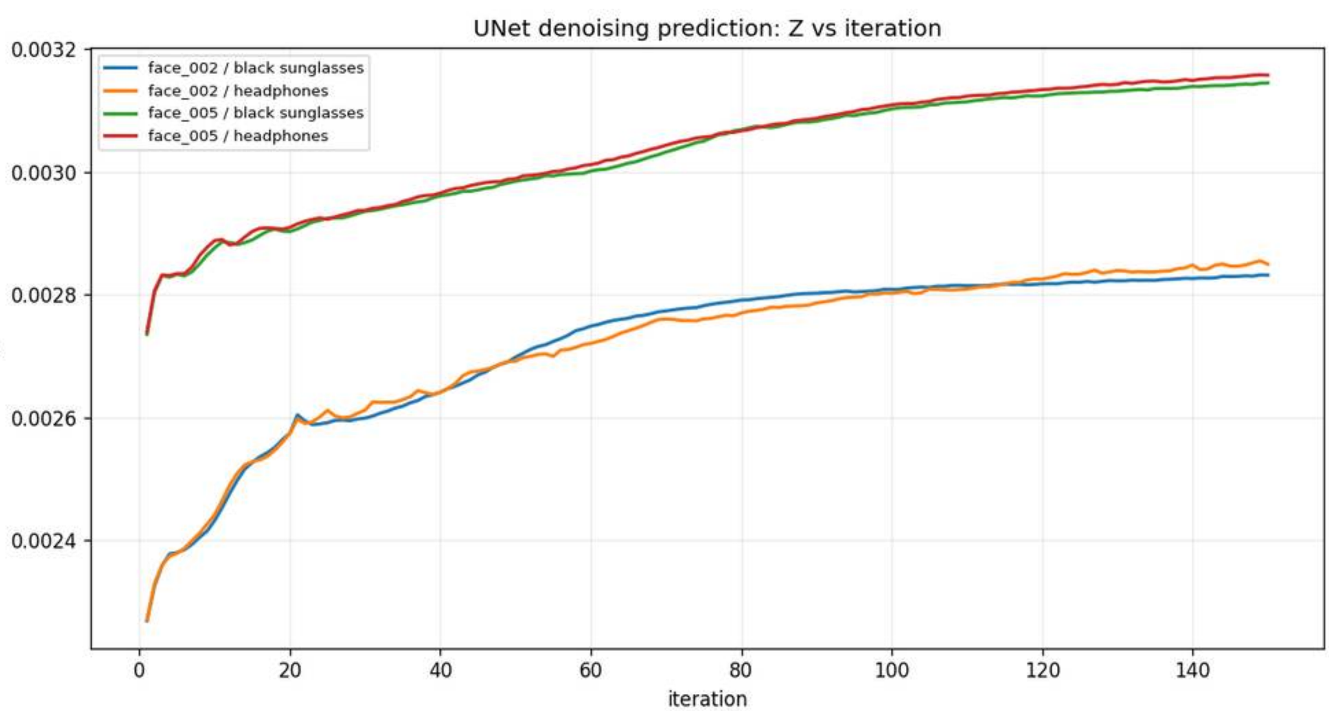


Geometry component contribution normalized

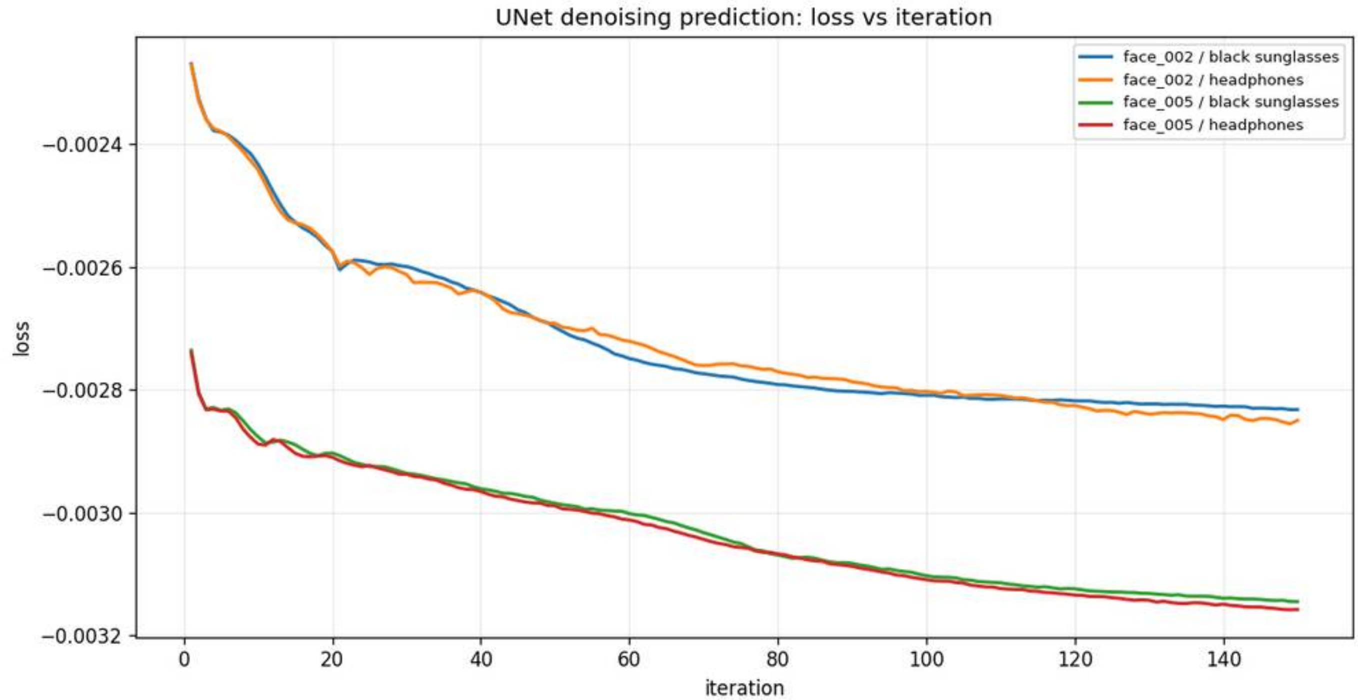


UNet denoising prediction

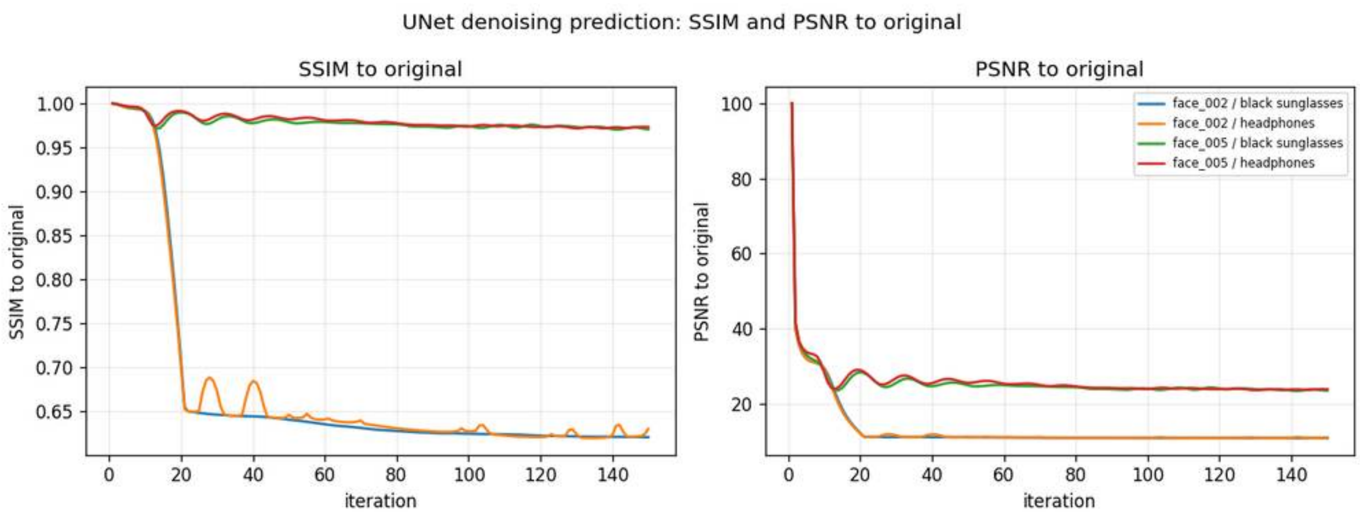
UNet denoising prediction: Z vs iteration



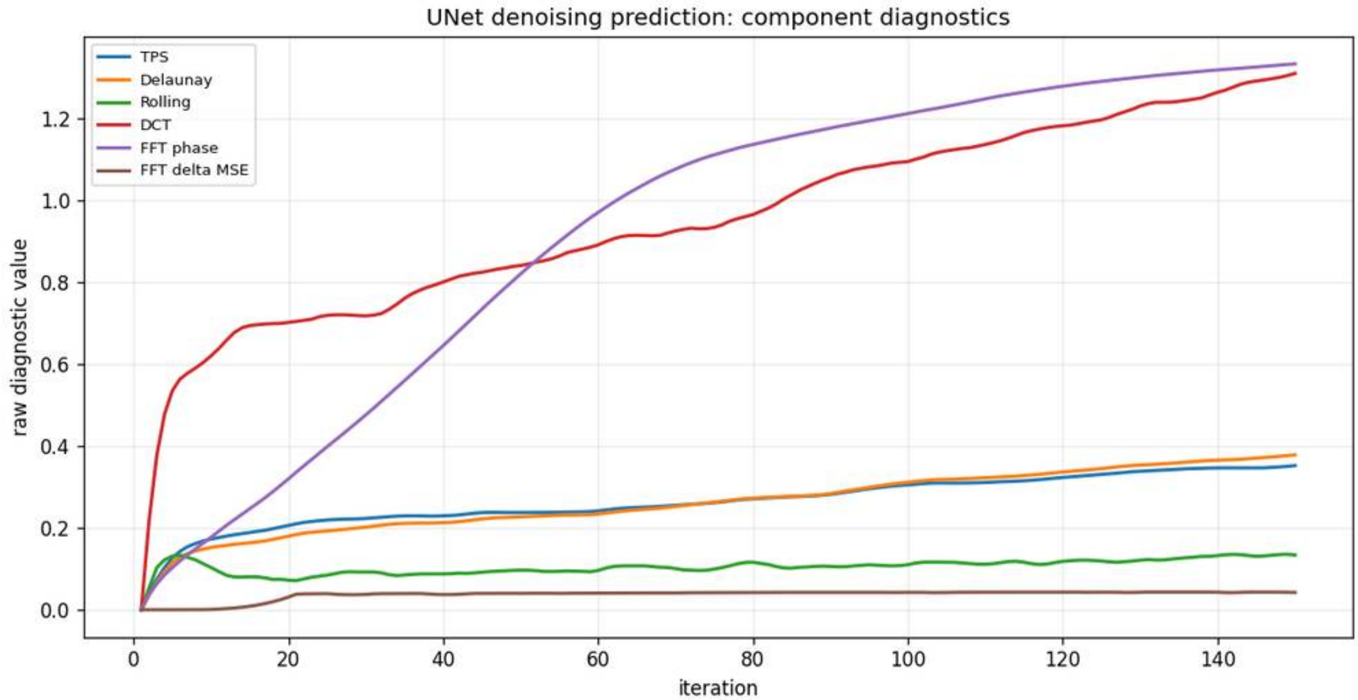
UNet denoising prediction: loss vs iteration



SSIM and PSNR to original



Geometry component contribution



Geometry component contribution normalized